INTRODUCTION TO INFORMATION GEOMETRY

Geodesics and Statistical Midpoints

Week 8 · Lecture 2

Christopher Lee

A constant-speed spherical geodesic

Map P and R to $s_0 = 2\sqrt{P}$ and $s_1 = 2\sqrt{R}$. Let $\alpha = \arccos B(P, R)$. Then

$$s(t) = \frac{\sin((1-t)\alpha)}{\sin\alpha} s_0 + \frac{\sin(t\alpha)}{\sin\alpha} s_1.$$

The probability path is $\gamma_i(t) = s_i(t)^2/4$.

Why the speed is constant

Choose an orthonormal basis e_0, e_1 for the plane of the great circle. Then the same path has the form

$$s(t) = 2\big(\cos(t\alpha)e_0 + \sin(t\alpha)e_1\big).$$

Differentiate:

$$\|s'(t)\| = 2\alpha.$$

Its length over $0 \le t \le 1$ is $2\alpha = d_F(P, R)$.

The Fisher midpoint formula

At $t = \frac{1}{2}$, both sine coefficients agree. The probability midpoint between P and R is

$$\gamma_i(\tfrac{1}{2}) = \frac{(\sqrt{p_i} + \sqrt{r_i})^2}{2(1 + B(P, R))}.$$

The squares make every coordinate nonnegative; the denominator makes them sum to one.

The Bernoulli midpoint

Find the constant-speed Fisher midpoint between $p = \frac{1}{4}$ and $p = \frac{3}{4}$. Recall $\theta(p) = 2 \arcsin \sqrt{p}$. The midpoint is $p_{\text{mid}} = ?$.

The Bernoulli midpoint

Find the constant-speed Fisher midpoint between $p = \frac{1}{4}$ and $p = \frac{3}{4}$. The arc coordinates are $\pi/3$ and $2\pi/3$. Their midpoint is $\pi/2$, so $p_{\mathrm{mid}} = \frac{1}{2}$.

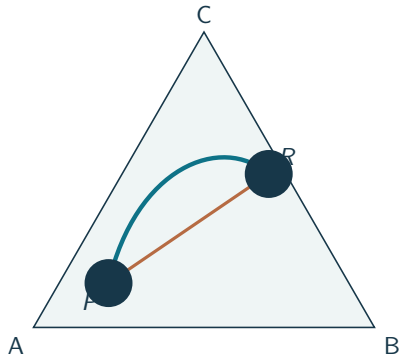
Mixtures trace straight probability segments

The ordinary mixture between distributions is

$$M(t) = (1 - t)P + tR.$$

It is a straight segment in the affine simplex drawing. The Fisher geodesic comes from a great-circle arc after taking square roots. For typical three-outcome endpoints, the two paths differ.

Two paths between the same laws



Schematic: copper shows a mixture segment; teal shows a curved Fisher route.

Hellinger distance and spherical chords

The square-root overlap also determines a Euclidean chord:

$$\|2\sqrt{P} - 2\sqrt{R}\|^2 = 8(1 - B(P, R)).$$

In one common normalization, Hellinger distance satisfies

$$h(P, R)^2 = 1 - B(P, R).$$

The chord and the spherical arc agree to first order for nearby laws.

Chord and arc between pure laws

Take $P = (1, 0)$ and $R = (0, 1)$. Their overlap is $B = 0$. Find

$$h(P, R), \quad \|2\sqrt{P} - 2\sqrt{R}\|, \quad d_F(P, R).$$

Chord and arc between pure laws

Take $P = (1, 0)$ and $R = (0, 1)$. Their overlap is $B = 0$, so

$$h(P, R) = 1, \qquad \|2\sqrt{P} - 2\sqrt{R}\| = 2\sqrt{2},$$

$$d_F(P, R) = \pi.$$

Global distance and the local metric agree

For a small tangent change h, with $\sum_i h_i = 0$,

$$B(P, P+h) = 1 - \frac{1}{8} \sum_i \frac{h_i^2}{p_i} + O(\|h\|^3).$$

It follows that

$$d_F(P, P+h)^2 = \sum_i \frac{h_i^2}{p_i} + O(\|h\|^3).$$

Curvature and the simplex boundary

The categorical Fisher manifold is an open region of the radius-2 sphere. Its sectional curvature is

$$K = \frac{1}{2^2} = \frac{1}{4}.$$

The positive interior is incomplete. Including distributions with zero coordinates gives its metric completion as the closed spherical orthant.

Two geometries of the same state space

The affine simplex represents mixtures as straight segments. The Fisher metric represents the same probability states as an orthant of a sphere. Next week we will use that metric to construct gradient flows on spaces of distributions.